

Error Detection and Correction: Catch the Corrupted Message

AIM

To detect transmission errors using parity bits in Python.

PROGRAM

```
msg = input("Enter message: ")
binary = "".join(format(ord(i), '08b') for i in msg)
ones = binary.count('1')
if ones % 2 == 0:
    parity = 0
else:
    parity = 1
print("Binary:", binary)
print("Parity Bit:", parity)
received = list(binary)
if received[0] == '0':
    received[0] = '1'
else:
    received[0] = '0'
received = "".join(received)
if received.count('1') % 2 == parity:
    print("OK")
else:
    print("ERROR DETECTED")
```

Output

Clear

Enter message: Hello
Binary: 0100100001100101011011000110110001101111
Parity Bit: 0
ERROR DETECTED

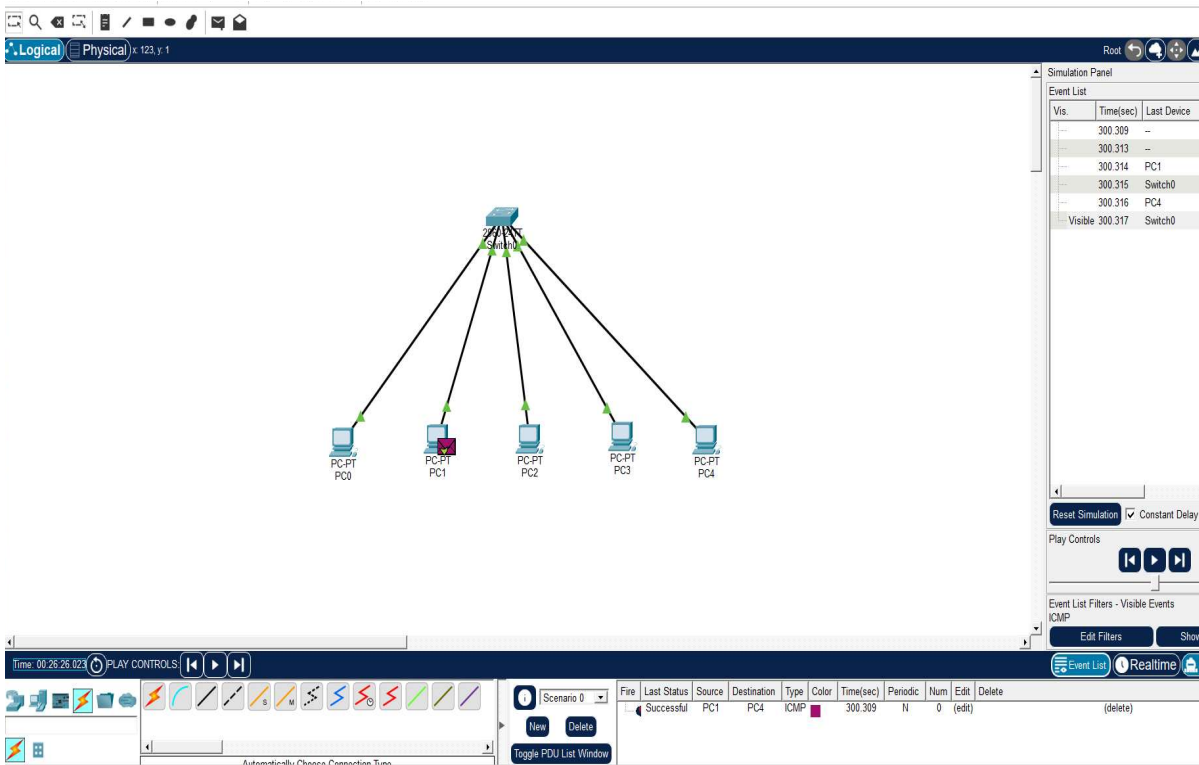
=== Code Execution Successful ===

Observation

The program converts a message into binary format and calculates a parity bit. A bit is intentionally changed to simulate transmission noise. The parity check detects the error and displays "ERROR DETECTED". This shows how parity bits help identify corrupted data during transmission.

LAN & Multiple Access

Topology



IP Configuration

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::206:2AFF:FE10::C1D4

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

Ping Test

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Observation

The network was created using 5 PCs connected to a switch. IP addresses were assigned to all PCs in the same network. Ping testing was performed

and communication between the PCs was successful. The switch forwards data only to the destination device, which reduces network traffic and collisions. This setup demonstrates efficient LAN communication using a switch.

Final Format

LAN & Multiple Access: Wi-Fi Channel Analysis

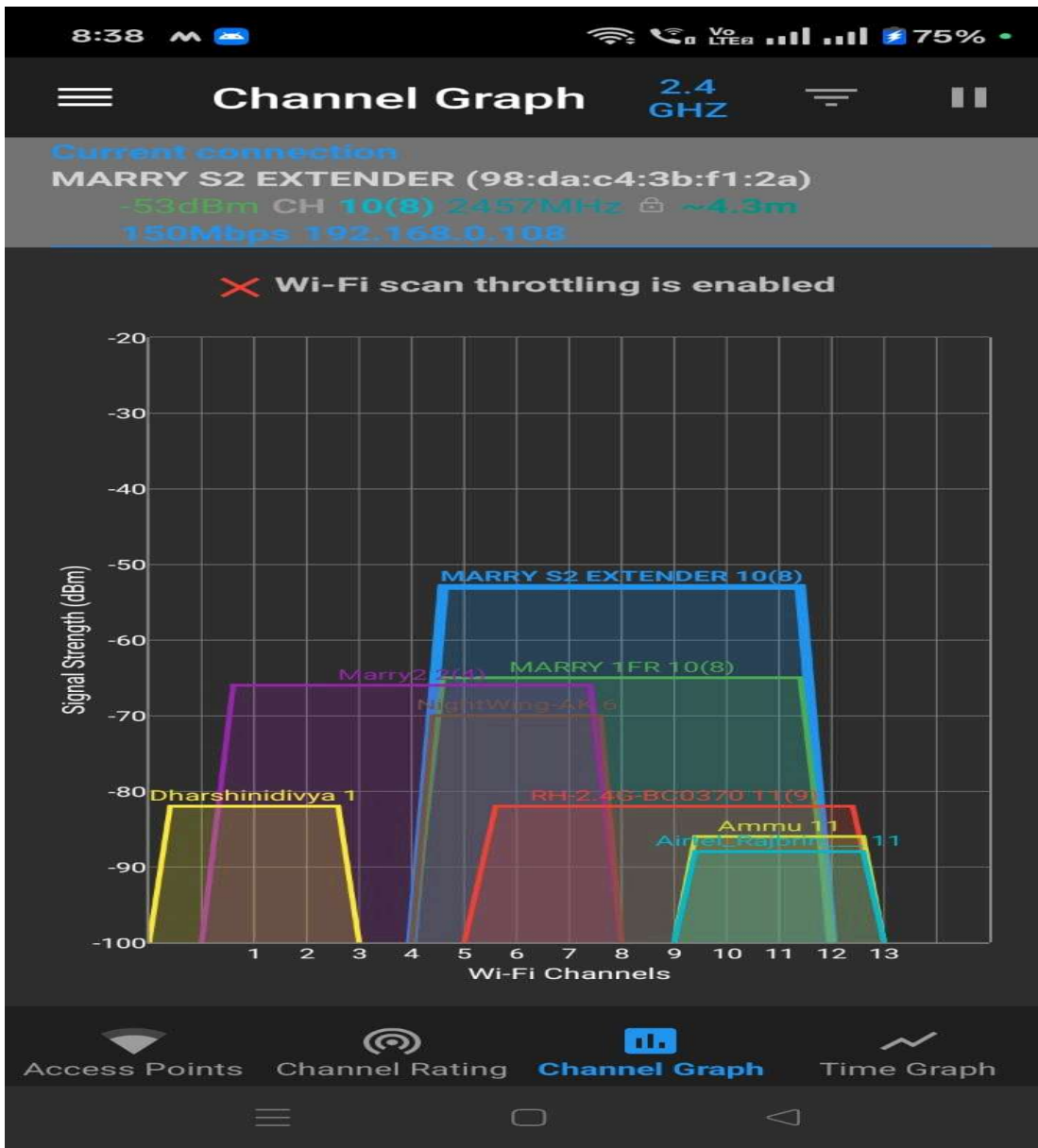


Table 1: Nearby Wi-Fi Networks and Channels

Wi-Fi Name	Channal
MARRY S2 EXTENDER	10
MARRY 1FR	10
Marry2	2
NightWing-AK	6
Dharshinidivya	1
Ammu	11

Observation:

- Channel 10 is used by MARRY S2 EXTENDER and MARRY 1FR.
- Channel 6 is used by NightWing-AK.
- Channel 1 is used by Dharshinidivya.
- Channel 11 is used by Ammu.
- Multiple networks share nearby channels, which may cause interference.